# CLASS 8 — SCIENCE · FORMULA SHEET

Chapter 8: **Nature of Matter: Elements, Compounds and Mixtures** | Quick revision, one page

## Classification of matter

- Pure substances: elements and compounds. Impure: mixtures.
- Element: one kind of atom; cannot be split by chemical methods (e.g. $O_2$, Fe, Na).
- Compound: two or more elements chemically combined in a fixed ratio (e.g. $H_2O$, $CO_2$, NaCl).
- Mixture: two or more substances not chemically combined, in any ratio; keep their properties.

## Symbols of elements

- Na (sodium), K (potassium), Fe (iron), Cu (copper), Ag (silver), Au (gold), Pb (lead), Hg (mercury), C, S, O, H, N.

## Metals and non-metals

- Metals: lustrous, malleable, ductile, sonorous, good conductors of heat and electricity (mercury is liquid).
- Non-metals: dull, brittle, poor conductors (exceptions: graphite conducts; iodine is lustrous).
- Sodium and potassium are stored under kerosene.
- Metal oxides are generally basic; non-metal oxides are generally acidic.

## Mixtures and separation

- Homogeneous (uniform): salt solution, air, alloys. Heterogeneous: sand and water, muddy water.
- Methods: handpicking, sieving, filtration, evaporation, magnetic separation, decantation, sedimentation.
- Alloys are mixtures of metals: brass (Cu + Zn), bronze (Cu + Sn).

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